A Markov Decision Process to Enhance Power System Operation Resilience during Hurricanes

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***Abstract*—This paper proposes a probabilistic proactive generation redispatch strategy to enhance operational resilience of power grids in preparation to and during hurricanes. Existing resilience enhancement strategies usually focus on reducing load curtailments during extreme events without proactively preparing power systems for predicted extreme events. Fragility of system components to extreme events and changing system conditions and constraints exhibit further computational burden and complexities to determine optimal system operation states. In this work, a multi-objective mixed integer linear programming optimization problem is formulated to minimize the amount of load curtailments and generation operation costs when a power system is impacted by a hurricane while adhering to system operational and technical constraints (e.g., ramping rates, minimum up/down times, line constraints, start-up/shut-down generation costs). A Markov decision process is used to model system state transitions and to provide generation redispatch strategies for each possible system state considering component failure probabilities, hurricane spatiotemporal properties, and load variations. The CPLEX solver is integrated with MATLAB to formulate and solve the optimization problem. The proposed strategy is demonstrated on several systems under various hurricane scenarios including the IEEE 30-bus system under various hurricane scenarios. The results show that the proposed solution can improvement power system resilience against hurricanes.**

***Index Terms*—Extreme weather event, generation redispatch, hurricanes, Markov decision process, resilience.**

I. Introduction

In the last decade, extreme weather events have impacted power system operation resulting in extended power outages and noticeable economic losses [1], [2]. In the United States, the estimated economic losses due to extreme weather-related outages is more than $25 billions per year [3]. Resilience enhancement strategies for power systems have been mainly based on mitigation and restoration approaches such as mobile energy storage devices, topology switching, and load shedding [4], [5]. However, proactive strategies against predicted extreme events, such as hurricanes, have not been given much attention. Also, various generation dynamic constraints, which playa vital role in resilience enhancement, have usually been neglected to reduce modeling complexities [6]. Impacts of load variations, future potential failures, and event attack time (i.e., the instant at which an extreme event hits the system) have been ignored resulting in less realistic models and results [1]. Thus, implementing a resilience enhancement strategy that account for operation cost, generation dynamic constraints, load variations, and extreme weather stochastic properties has become more important than ever before.

Resilience enhancement strategies vary based on the study period into: prior to, during, and after an event [7]. Operational resilience enhancement focuses on providing an immediate solution to reduce the impacts of adverse events via utilization of the available assets. In, [8] a procurement plan with a minimal cost has been proposed to ensure the availability of sufficient black start resources prior to an event. However, spatiotemporal characteristics of extreme events have been neglected. Various resilience enhancement strategies have also been proposed including maintenance planning [9] and mobile energy storage allocation to prepare the system before an event. In [10], a proactive resilience enhancement strategy based on generation redispatch has been proposed to reduce the cost of load curtailments during hurricanes where operational costs as well as load variations have been ignored. Authors of [6] have proposed a deterministic methodology to reduce the amount of load curtailments and operational costs for a day-head interval with hurricane events. However, the approach presented in [6] neglects the stochastic behavior of component failures.

This paper proposes a generation redispatch strategy to enhance the operational resilience of power grids against hurricanes. Markov decision process (MDP) is used to capture the stochastic behavior of component failures to determine the optimal generation dispatch as a hurricane evolves. A Markov state represents a specific system topology based on the available components. The proposed strategy minimizes the amount of load curtailments and generator operational costs during an event. Several system dynamic constraints have been considered including transmission capacity limits and generation limits (e.g., ramping rates, and up/down times). To capture load variations, historical load profiles are used where the hurricane is assumed to take place during peak load hours. A mixed integer linear programming optimization problem is formulated using the recursive Markov process on MATLAB environment integrated with CPLEX solver to solve for optimal generation redispatch strategies. The proposed method is implemented on the IEEE 30-bus transmission system. Analysis and validation of the results are also provided.

The rest of the paper is organized as follows. Section II describes the proactive generation redispatch strategy. Section III formulates the Markov decision process to minimize load curtailments and overall operation costs during extreme weather events. Section IV illustrates the proposed approach on the IEEE 30-bus system and discusses the results. Section V provides concluding remarks.

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II. Proactive Generation Redispatch

This section describes the proposed resilience enhancement strategy for transmission systems against hurricanes. First, it describes the impacts of propagation of hurricane on power grid components. Then, it explains Markov decision process to formulate a probabilistic generation redispatch algorithm.

A. Impacts of Hurricane Propagation

Each extreme weather event has a unique propagation properties and spatiotemporal characteristics that directly impact the performance of system components. Proper probabilistic models are usually used to identify potentially impacted components and their probability of failure. Although most resilience-based studies have used probabilistic fragility curves to model the stochastic behavior of system components against extreme weather events [11], other studies have simulated predefined failure scenarios based on forecasting models [6]. Due to the temporal and geographical progression of hurricanes, various components in the system can be impacted at sequential time intervals [10]. Fig. 1 shows a scenario where three system components are on the trajectory of a hurricane. Failed components may not be restored during hurricanes since they are fast acting weather events.

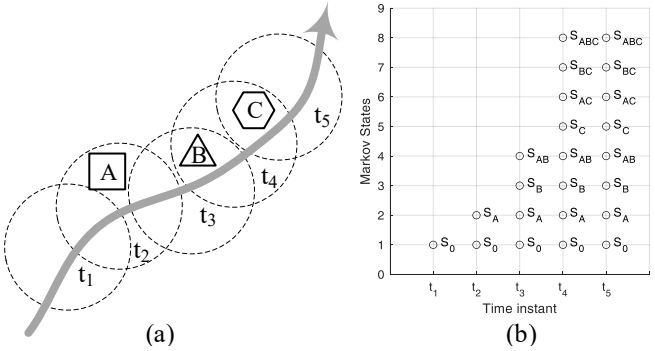


Fig. 1. Components on the trajectory of a hurricane

B. System States during Hurricane

Due to the sequential failure behavior introduced by hurricanes, the power grid might have different operating states at each time instant. A Markov state can be defined to represent a unique system topology based on the available components. Total number of Markov states is $2^{N_{c,t}}$ where $N_{c,t}$ is the total number of impacted components at time t. Assuming that failed components are not recovered during the hurricane period, the set of impacted components at time t lists the current and previously impacted components. Fig. 1 describes the propagation of component failures on Markov states during hurricane. S_o represents Markov state with no component failure, whereas S_{ABC} denotes Markov state where all components are in failure state. Utilizing Markov process to represent the behavior of power system from one time instant to another requires evaluating transition probabilities from one state to another. The transition probability from state $S_{i,t}$ to $S_{i',t+1}$ can be evaluated as follows.

$$P(S_{i,t}, S_{i',t+1}) = \prod_{m \in \Omega_{C,t+1}} P(o_{m,t}, o_{m,t+1}) \ , i \in \Omega_{S,t} \ , \quad (1)$$

$$P(o_{m,t}, o_{m,t+1}) = \begin{cases} 1 & o_{m,t} = 0, o_{m,t+1} = 0 \\ 0 & o_{m,t} = 0, o_{m,t+1} = 1 \\ 1 - \lambda_{m,t+1} & o_{m,t} = 1, o_{m,t+1} = 1 \\ \lambda_{m,t+1} & o_{m,t} = 1, o_{m,t+1} = 0 \end{cases} \quad (2)$$

where $\Omega_{S,t}$ is the set of all system states at a given time t; $\Omega_{C,t+1}$ is the set of all impacted components at time $t+1$; $\lambda_{m,t+1}$ is the failure probability of component m at time $t+1$; and $o_{m,t} = 0$ and $o_{m,t+1}$ are the statuses of the component m at times t and $t+1$, respectively.

C. Recursive Markov Process

Sequential failure of power system components results in noticeable changes in the performance of power grids. During extreme weather events, the priority should be given to the reduction of load curtailments rather than operational costs to maintain grid resilience. Also, the stochastic future behavior of a hurricane on system components should be considered in order to provide a feasible strategy that satisfies all system constraints such as ramping rates, minimum up and down times, and varying load demand. Proactive generation redispatch aims to determine the optimal generation levels of each operating generator for a specific period of time given the current and predicted future system states. Assurance of assets availability, such as generating units and transmission lines, during and after a hurricane is a vital constraint to maintain resilient operation of the system. In severe situations, the power grid could be islanded into multiple grids where the generation level at each islanded grid should stably supply its loads. Moreover, it is preferable to restore curtailed loads in a fast, efficient, and economical way to improve the overall operational resilience level of power system.

Uncertainties of component failures during hurricanes impose additional burden for system operators to determine the best decision, according to current and future system states. Each decision impacts the overall performance of the system during the entire period of an event. Since generation dispatch usually takes place in terms of minutes, a discrete-time Markov decision process can be used to model the whole process. The MDP determines the optimal action (decision) at each time instant based on current system states as well as possible future states. The backward induction method and the value iteration method [12] have been used to find solution for each state in MDP. However, when time-dependent constraints correlating Markov states at sequential time instants are considered, linear scalarization method can be used to transform the multi-objective optimization problem into a single objective optimization problem [10].

III. Markov Decision Process Formulation

This section explains the formulation of the multi-objective optimization problem to minimize the overall operating costs and load curtailments. To consider the sequential reconfiguration of system network during hurricane events, a recursive model based on discrete-time MDP is developed. Various system constraints and hurricane spatiotemporal properties are considered in the proposed algorithm.

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A. Objective Function

The optimal generation redispatch strategy for a specific system state $S_{i,t}$ at a given time t is expressed as follows.

$$v_t^*(S_{i,t}) = \min\{v_t(S_{i,t}, A_{a,t}), a \in \Omega^A\}, i \in \Omega_{S,t}, t \in \Omega^T \quad (3)$$

where $A_{a,t}$ denotes a set of actions, i.e., generators output power; $v_t(S_{i,t}, A_{a,t})$ is the expected overall cost for state $S_{i,t}$ under a specific action $A_{a,t}$; Ω^A represents the set of all possible actions; $\Omega_{S,t}$ is the set of all system states at a given time instant t; and Ω^T is the set of all time instants.

The value of each state in MDP can be evaluated as follows.

$$v_t(S_{i,t}, A_{a,t}) = C_t(S_{i,t}, A_{a,t}) \; + \sum_{i' \in \Omega^S_{i,t+1}} [P(S_{i,t}, S_{i',t+1}) \; . \; v_{t+1}(S_{i',t+1}, A_{a',t+1})], \quad (4)$$

In (4), $\{a, a'\} \in \Omega^A$, $i \in \Omega_{S,t}$, and $\{t, t+1\} \in \Omega^T$ where $P(S_{i,t}, S_{i',t+1})$ is the transition probability from state $S_{i,t}$ to all possible states in the proceeding time instants; and $C_t(S_{i,t}, A_{a,t})$ represents the immediate cost of state $S_{i,t}$ given action $A_{a,t}$, which can be expressed as follows.

$$C_t(S_{i,t}, A_{a,t}) = W_1 \; . \; C_{cu} \; . \sum_{n \in \Omega^N} Cu_{n,t,i} \; + W_2[\sum_{j \in \Omega^G} C_f(P^G_{j,t,i}) + C_{su}(T^{ON}_{j,t,i}) + C_{sd}(T^{OFF}_{j,t,i})] \quad (5)$$

where $Cu_{n,t,i}$ represents the amount of load curtailment at node n; C_{cu} is the cost of curtailed loads; $C_f(P^G_{j,t,i})$ is the fuel cost of generator j; Ω^N represents the set of all buses; Ω^G represents the set of all generators; and $C_{su}(T^{ON}_{j,t,i})$ and $C_{sd}(T^{OFF}_{j,t,i})$ are the startup and shutdown cost of generator j, respectively.

B. Constraints

Several constraints are considered as follows.

1) Power Balance:

The power balance of system state $S_{i,t}$ at time t can be expressed as follows.

$$\sum_{j \in \Omega^G_n} P^G_{j,t,i} - (L_{n,t,i} - Cu_{n,t,i}) + \sum_{n' \in \Omega^N_n} P^L_{n,n',t,i} = 0, \forall n \in \Omega^N \quad (6)$$

where Ω^G_n represents the set of generators connected to bus n; $P^G_{j,t,i}$ is the jth generator real power at bus n; $L_{n,t,i}$ is the amount of load in MW at bus n; $Cu_{n,t,i}$ is the amount of load curtailed at bus n; Ω^N_n represents the set of all buses connected to bus n; and $P^L_{n,n',t,i}$ represents the power flow from bus n to bus n' at time t.

2) Transmission Flow Limits:

The power flow through a specific line connected at bus n of system state $S_{i,t}$ can be defined as follows.

$$B_{n,n'}.(\theta_{n,t,i} - \theta_{n',t,i}) - P^L_{n,n',t,i} \leq P^{Max}_{n,n',t,i} \; \forall n \in \Omega^N \quad (7)$$

$$B_{n,n'}.(\theta_{n,t,i} - \theta_{n',t,i}) - P^L_{n,n',t,i} \geq P^{Min}_{n,n',t,i} \; \forall n \in \Omega^N \quad (8)$$

where $B_{n,n'}$ represents the susceptance of the line connecting nodes n and n'; $\theta_{n,t,i}$ and $\theta_{n',t,i}$ are the voltage angles at buses n and n', respectively; and $P^{Max}_{n,n',t,i}$ and $P^{Min}_{n,n',t,i}$ are the maximum and minimum line flow ratings, respectively.

3) Load Curtailment Limits:

For each $S_{i,t}$ during a hurricane, the amount of load curtailment at each bus should be less than or equal the total amount of load at the same bus as follows:

$$0 \leq Cu_{n,t,i} \leq L_{n,t,i} \; \forall n \in \Omega^N, \; \forall t \in \Omega^T \quad (9)$$

4) Ramping Rates of Generating Units:

The ramping rates of each generator should be satisfied as follows.

$$P^G_{j,t+1,i} - P^G_{j,t,i'} \leq (2 - u_{j,t,i} - u_{j,t+1,i'}).P^{G,Min}_j + (1 + u_{j,t,i} - u_{j,t+1,i'}).R^{UP}_j \; \forall i' \in \Omega^S_{i,t+1} \quad (10)$$

$$P^G_{j,t,i} - P^G_{j,t+1,i'} \leq (2 - u_{j,t,i} - u_{j,t+1,i'}).P^{G,Min}_j + (1 - u_{j,t,i} + u_{j,t+1,i'}).R^{DN}_j \; \forall i' \in \Omega^S_{i,t+1} \quad (11)$$

where for the jth generator, $P^G_{j,t+1,i'}$ is generated power in state $S_{i',t+1}$; $u_{j,t,i}$ and $u_{j,t+1,i'}$ are generator statuses at states $S_{i,t}$ and $S_{i',t+1}$, respectively; $P^{G,Min}_j$ and $P^{G,Max}_j$ are the minimum and maximum generation power; and R^{UP}_j and R^{DN}_j are the up and down ramping rates, respectively.

5) Generators Minimum Up/Down Time:

Since the proactive redispatch is time-dependent, minimum up and down times for each generator should be satisfied as follows.

$$\sum_{t-UT+1}^{t} T^{ON}_{j,t,i} \leq u_{j,t,i''} \; \forall t \in \{UT, \cdots, T\} \quad (12)$$

$$\sum_{t-DT+1}^{t} T^{OFF}_{j,t,i} \leq 1 - u_{j,t,i''} \; \forall t \in \{DT, \cdots, T\}, \; \forall j \in \Omega^G, \; \forall i'' \in \Omega^S_{i,t+} \quad (13)$$

where $T^{ON}_{j,t,i}$ and $T^{OFF}_{j,t,i}$ are the turn on/off signals of jth generator at state $S_{i,t}$, respectively; UT and DT are the minimum up/down times for same generator, respectively; and $\Omega^S_{i,t+}$ is the set of all possible transition states starting at state $S_{i,t}$.

6) Power Limits of Generating Units:

The generated power of each generator can be as expressed as follows.

$$P^{G,Min}_j.u_{j,t,i} \leq P^G_{j,t,i} \leq P^{G,Max}_j.u_{j,t,i} \; \forall j \in \Omega^G \quad (14)$$

where $P^{G,Min}_j$ and $P^{G,Max}_j$ are the lower and upper limits of jth generator, respectively; and Ω^G is the set of all generators.

7) Generator Status:

The status of each generator at state $S_{i,t}$ is represented by a binary number as follows.

$$u_{j,t,i} \in \{0, 1\} \; \forall j \in \Omega^G \quad (15)$$

8) Voltage Angle Limits:

Voltage angle at bus n at state $S_{i,t}$ can be expressed as follows.

$$\theta^{Min}_n \leq \theta_{n,t,i} \leq \theta^{Max}_n \; \forall n \in \Omega^N \quad (16)$$

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IV. Implementation and Results

The proposed approach is applied on the IEEE 30-bus system. The MDP is formulated using CPLEX solver integrated with MATLAB. The hurricane is assumed to pass through the system as shown in Fig. 2 and to last for 25 minutes sampled in set of 5 minutes. At each time instant, components may fail due to the spatiotemporal properties of the hurricane. The set of impacted components and their failure probabilities are given in Table I, using the approach given in [10].

TABLE I
List of Impacted Components with their Probability of Failure

Time Instant	Component No.	Description	Failure Probability
t_1	–	–	–
t_2	C_1 C_2	Line 15-23 Line 18-19	0.25 0.22
t_3	C_4	Line 16-17	0.2
t_4	C_6 C_7	G_6 Line 4-6	0.08 0.18
t_5	C_8 C_9 C_{10}	G_2 Line 2-6 Line 2-5	0.08 0.15 0.12

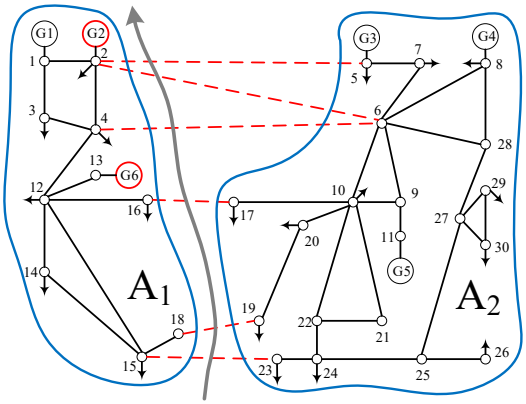


Fig. 2. Hurricane propagation on IEEE 30-bus system

Several test cases are simulated to validate the accuracy and effectiveness of the proposed method. The impact of load variations is considered by scaling the system nominal load based on load demand profile obtained from [13]. The hurricane event is assumed to take place during load peak time to create more severe circumstances. On the other hand, proper scaling weights are used to prioritize the cost of load curtailment over the operational costs. Two simulation cases are considered to assess the sensitivity between the initial generation level and algorithm performance.

(1) Predefined Initial Generation Level: In this case, initial generation levels are obtained by solving optimal power flow under normal operating conditions. The calculated initial generation profile is integrated into the optimization problem to determine the optimal generation redispatch scenarios. Fig. 3 shows the optimal generation strategy for three scenarios compared with the generation dispatch under normal operation conditions. The three scenarios are selected as follows: (a) no component fail, (b) only transmission lines fail, and (c) all components on the hurricane path fail. The overall cost of each redispatch strategy is evaluated and compared with normal operation cost as illustrated in Fig. 5.

It is noticeable that the MDP takes in consideration the potential future failures of system components. Both G_1 and G_2 are ramping down in all studied redispatch scenarios to shift the reliance of the system on the largest generation unit. Although G_3, G_4, and G_5 are ramping up to compensate for the reduction of generation from G_1 and G_2 as well as achieve the increasing demand, the net ramp up/down of all generators is not sufficient to supply all load demands resulting in some load curtailments. On the other hand, G_6 does not continue to ramp up for the whole period because the possibility to fail at t_4. Although system components do not fail in scenario (a), load is curtailed at earlier time instants to avoid much larger curtailment in proceeding instants. Also scenario (b) describes the generation portfolio when islanding occurs at t_5 which shows further curtailment due to insufficient power for each region. The total amount of load curtailment increases in scenario (c) due to the loss of G_2. The impact of the generation level prior to the event on the performance of the redispatch strategy has been noted clearly in scenario (a) and (b). For instant, if G_2 was producing much less power, it would have behaved as G_6 and shut down earlier in time.

(2) Unknown Initial Generation Level: Due to the sensitivity of generation redispatch and ramping to initial generation levels, MDP algorithm is used to determine the optimal initial generation level as well as the dispatch profile to reduce the amount of load curtailment. Fig. 4 shows the optimal initial generation level and redispatch strategy for scenarios (b) and (c) from the previous case. Also, the overall cost is compared with results of the previous case as illustrated in Fig. 5.

The obtained generation dispatch strategies for scenario (b) and (c) are significantly different than previous results in many ways. All generation units are initialized to deliver 40% to 70% of their maximum capacity resulting in higher initial generation of G_3, G_4, and G_5. Also, the initial generation level of G_2 has been reduced to be able to completely shut down by reaching t_5. The MDP algorithm utilizes G_2 as long as possible since it has lower overall operating costs than other units. The obtained strategies for scenarios (b) and (c) are identical reflecting the effectiveness of the MDP algorithm to consider future potential generation outage. The ramping of G_3, G_4, and G_5 is mainly due to increase in load demand. G_6 reaches a complete shutdown state at t_3 before it is potentially impacted at t_4. Although a significant reduction in load curtailment profile is obtained, both scenarios encounter small amount of load curtailment at the last time instant t_5.

Fig. 5 describes the variation of cost values for all scenarios under various constraints. Three feasible initial generation conditions are used to assess the performance of the redispatch algorithm: (1) 40% to 70% of maximum generation capacity; (2) 100% to 230% of minimum generation capacity; and (3) minimum to maximum generation capacity. Condition (2) imposes more curtailment costs than condition (1) and (3) because generators are initialized at low levels. When the initial generation level constraint is relaxed, the algorithm reaches very close total cost compared to normal operation costs in case of scenario (a).

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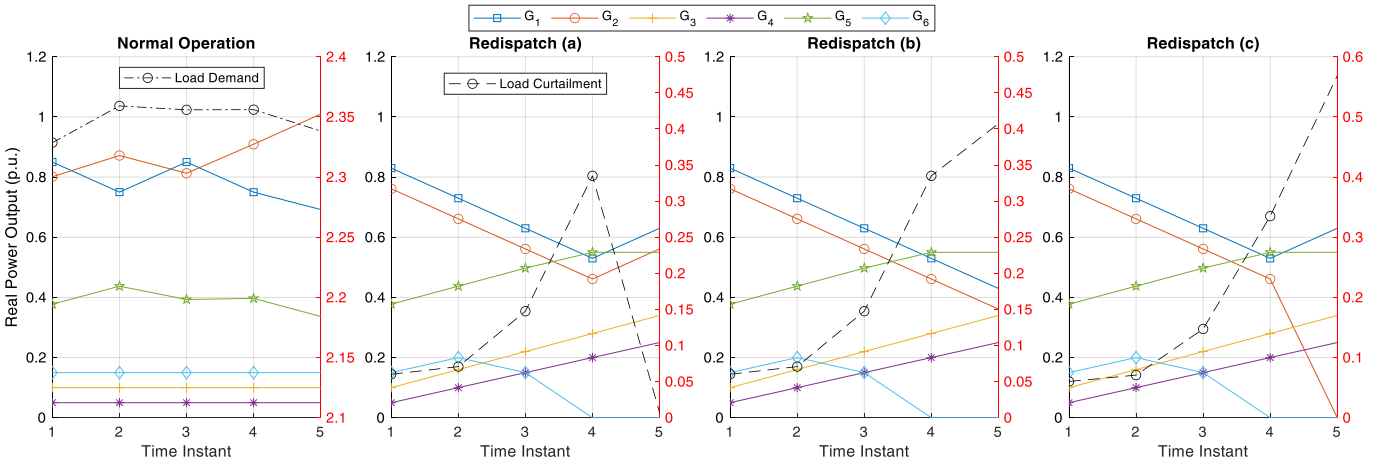


Fig. 3. Optimal Strategies under Predefined Initial Generation Level

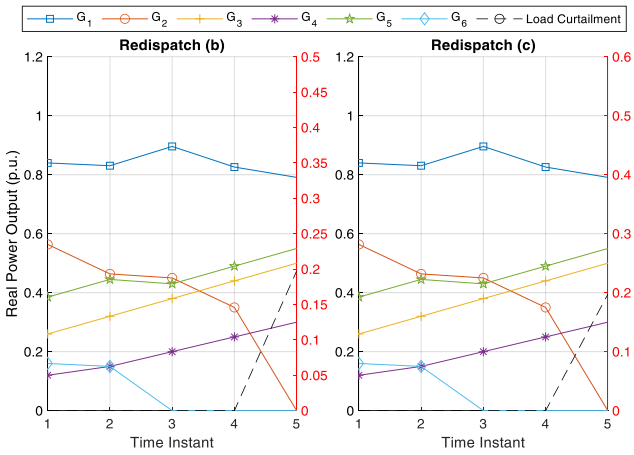


Fig. 4. Optimal Strategies under Unknown Initial Generation Level

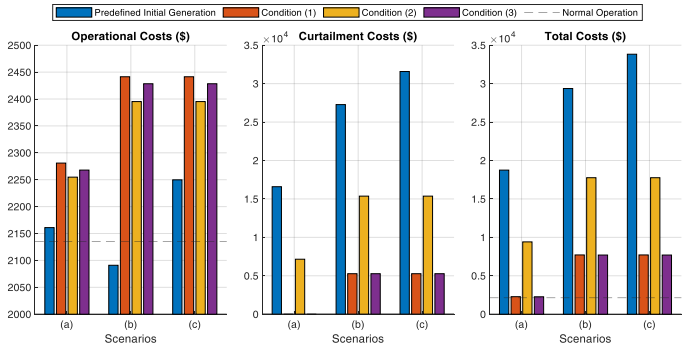


Fig. 5. Cost Analysis

V. Conclusion

This paper has proposed a probabilistic proactive generation redispatch strategy to enhance the operation resilience of power grids during hurricanes. The proposed method minimizes load curtailments as well as system operational under various system constraints. MDP was utilized to encounter the impact of failure probabilities for the optimal redispatch strategy. The proposed method was demonstrated on the IEEE 30-bus system. The results showed that generation redisptatch strategy enhances the operational resilience of power grids. It also showed the effectiveness of the algorithm to reduce/eliminate the overall costs via proactive redispatch. The algorithm considers the current and potential failures of system components to reduce the amount of load curtailment.

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